

Assessments

In the following pages, we will review all the practice questions from each of the chapters in this book and provide the correct answers.

Chapter 1 – From Cloud to Cloud Native and Kubernetes

1. B, D
2. C
3. B, D
4. D
5. B, D
6. B, C, D
7. A
8. C
9. A, B, D
10. A
11. A, B
12. C
13. B, C, D
14. A, B, D
15. C

Chapter 2 – Overview of CNCF and Kubernetes Certifications

1. A, C, D
2. D
3. B, C, D
4. C
5. A, B, D
6. A, B

7. A, B, C
8. D
9. A, B
10. C
11. C, D
12. B, C
13. A
14. A, B
15. B, C, D

Chapter 3 – Getting Started with Containers

1. A, C
2. A, C
3. D
4. A, D
5. C, D
6. A, C
7. B, D
8. A, C
9. B, D
10. A

Chapter 4 – Exploring Container Runtimes, Interfaces, and Service Meshes

1. D
2. B, C, D
3. B
4. D
5. D
6. B
7. C

8. C
9. B
10. D
11. A, B, C
12. B
13. A, D
14. B, C
15. A, C
16. A, C
17. B
18. D

Chapter 5 – Orchestrating Containers with Kubernetes

1. C
2. B, C, D
3. B, C
4. C, D
5. D
6. D
7. B
8. C
9. B, D
10. D
11. B, D
12. B, D
13. A, D
14. D
15. B
16. C
17. C
18. B
19. C

- 20. C
- 21. C, D

Chapter 6 – Deploying and Scaling Applications with Kubernetes

- 1. C, D
- 2. B, C
- 3. B
- 4. D
- 5. D
- 6. B
- 7. C
- 8. C
- 9. B
- 10. D
- 11. C
- 12. B
- 13. D
- 14. B, C
- 15. A
- 16. C
- 17. B
- 18. D

Chapter 7 – Application Placement and Debugging with Kubernetes

- 1. C, D
- 2. B
- 3. B
- 4. B
- 5. C

6. D
7. C
8. C
9. B
10. D
11. B
12. B, D
13. A
14. C
15. A
16. C

Chapter 8 – Following Kubernetes Best Practices

1. C
2. D
3. B
4. C
5. B
6. C
7. B
8. A
9. D
10. D
11. B, D
12. B
13. A
14. C, D
15. B

Chapter 9 – Understanding Cloud Native Architectures

1. C
2. B
3. A
4. D
5. B
6. D
7. C, D
8. C, D
9. A
10. D
11. B
12. D
13. A
14. A, C
15. A
16. D

Chapter 10 – Implementing Telemetry and Observability in the Cloud

1. C, D
2. B
3. D
4. B
5. D
6. B
7. C
8. C, D
9. B
10. D
11. A

- 12. D
- 13. A
- 14. C
- 15. D
- 16. A, C, D
- 17. D
- 18. C

Chapter 11 – Automating Cloud Native Application Delivery

- 1. C
- 2. A, B
- 3. B, D
- 4. C
- 5. A, B, D
- 6. B, C, D
- 7. C
- 8. D
- 9. C
- 10. B, D
- 11. A, C
- 12. C, D
- 13. A
- 14. C
- 15. C
- 16. A, C
- 17. A, C, D

Chapter 12 – Practicing for the KCNA Exam

Mock exam A

1. C
2. B
3. D
4. D
5. B
6. D. Deployment is the best option because ReplicaSet does not allow rolling updates and StatefulSet is not required for stateless applications.
7. C
8. D
9. A
10. D
11. B
12. D
13. A. DaemonSet is the best option because it can ensure a replica is running on each Kubernetes node.
14. C
15. A, C
16. C
17. D. Containers in one pod can file system mounts and communicate over localhost. All containers of one pod will always run together on the same node.
18. D
19. B
20. D. CI/CD is the correct answer because we need build and test automation and we don't need automatic deployments. CD is Continuous Delivery in this case.
21. A, C
22. D
23. B. Using namespaces is the best option as it allows to further restrict access by implementing RBAC policies. A team can be restricted to only one namespace.
24. A
25. B

- 26. B
- 27. D
- 28. A, B, C
- 29. B
- 30. D
- 31. C, D
- 32. B
- 33. D. HPA is the correct answer because we need to scale an application in the context of the question. Obviously, a cluster autoscaler would be required eventually to adjust the number of nodes.
- 34. B
- 35. C, D
- 36. B, D
- 37. B
- 38. B, C
- 39. D
- 40. B
- 41. D
- 42. B
- 43. A
- 44. B
- 45. B, D
- 46. A
- 47. B
- 48. D
- 49. A
- 50. C. Other options might either impact the stability or make it harder to operate the applications.
- 51. B
- 52. D
- 53. C
- 54. B
- 55. C

- 56. A, C
- 57. C. At the time of writing this book, it is approximately 3 releases per year.
- 58. D
- 59. C
- 60. B

Mock exam B

- 1. D
- 2. A, B
- 3. C
- 4. A
- 5. B
- 6. A
- 7. B
- 8. D
- 9. B
- 10. B
- 11. C
- 12. A, D
- 13. D
- 14. B, C, D
- 15. B, C
- 16. A, C
- 17. D
- 18. A, B, C. By default, kubectrl should work with one version above and one version below.
- 19. D
- 20. A
- 21. A
- 22. C
- 23. B
- 24. C
- 25. C, D

- 26. B
- 27. A
- 28. A
- 29. D
- 30. D
- 31. B
- 32. D
- 33. B
- 34. C
- 35. A
- 36. A
- 37. D
- 38. A, C
- 39. A
- 40. B
- 41. A
- 42. B
- 43. B, D
- 44. D
- 45. B
- 46. D. Deployment is the best option because it satisfies all the requirements, and the application is stateless.
- 47. B
- 48. A
- 49. A, C
- 50. C
- 51. B
- 52. D. Kubernetes namespaces allow configuration of resource quotas.
- 53. A
- 54. B
- 55. C

- 56. C, D. GitOps would be helpful for continuous deployment and FluxCD is a GitOps tool, so both A and B are incorrect answers.
- 57. C, D
- 58. B
- 59. D
- 60. C